



# University of Narowal

Department of Computer Science

Machine Learning - CSC 351

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## Climate Change And Global Warming Project

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# Climate Change & Global Warming Project

## Abstract

This report covers a machine learning study of global climate change and warming, built on two publicly available datasets: the Berkeley Earth Surface Temperature (BEST) dataset and the Our World in Data (OWID) CO2 Emissions dataset. The idea was simple to state but harder to do properly. We wanted a model trained on 265 years of temperature and emissions data that could actually survive scrutiny, not just post a good score. Along the way we also wanted to know which parts of the world are heating up fastest, and to put together a defensible forecast for global land temperature in 2050.

We followed a fairly cautious process. Four raw CSV files were cleaned one at a time instead of forced through a single pipeline. Features were engineered from 1,992 monthly records. Five candidate models were compared against each other before we settled on one. The first attempt, a Random Forest Regressor, returned an  $R^2$  of 0.9902 on the surface. That number looked great until we checked what was actually driving it. Month alone accounted for over 95 percent of the model's feature importance, meaning it had basically learned the calendar instead of the climate. It also could not extrapolate past 2015. We dropped it.

The model we kept was a Ridge Regression trained on six engineered features. It reached a test  $R^2$  of 0.9927, a mean absolute error of 0.2805 degrees Celsius, and a cross-validation standard deviation of 0.0005. Running it forward month by month to 2050 gave an absolute land temperature of 10.18 degrees Celsius. That works out to roughly 2.28 degrees of warming since 1850, which sits comfortably inside the range the IPCC has already projected.

On the geographic side, Mongolia came out on top at +1.836 degrees Celsius, with Kazakhstan and Kyrgyzstan close behind at +1.464 and +1.423 degrees. Ulan Ude and Chita led the city-level rankings. Taken together, these results support the idea that CO2-linked features are not just statistically convenient. They line up with the physical mechanism that is actually driving the warming.

## 1. Introduction

### 1.1 Background

Climate change is one of the defining problems of our time, and most people already know that in some general sense. What's less obvious is how steadily the numbers have been climbing. Global surface temperature has been rising for well over a century, driven mostly by human greenhouse gas emissions. The consequences are not abstract anymore. Sea levels are rising. Weather extremes

are becoming more common. Agriculture and public health are both feeling the effects in different ways.

The IPCC has documented this warming trend going back to the start of the industrial era in the mid-1800s, and CO<sub>2</sub> keeps showing up as the dominant driver behind it. Machine learning gives us a practical way to actually dig into datasets this large and pull something predictive out of them, rather than just describing the trend qualitatively. That's the motivation behind this project: apply supervised regression to 265 years of temperature and CO<sub>2</sub> data, quantify the relationship properly, find out where warming is hitting hardest, and build a 2050 forecast that is grounded in the data rather than guesswork.

## 1.2 Problem Statement

Turning raw climate records into something you can actually forecast with is not as simple as it looks on paper. A model can post an impressive R<sup>2</sup> and still be scientifically empty. It might be picking up on seasonal noise rather than the long-term trend. Or the train-test split might be leaking future information into training without anyone noticing until the numbers look too good to be true. We built this project around four questions:

- How has global land average temperature changed between 1750 and 2015?
- What is the statistical relationship between CO<sub>2</sub> emissions and global temperature?
- Which machine learning model is best suited to forecasting out to 2050?
- Which regions are warming fastest, and what does that mean for the 2050 forecast?

## 1.3 Objectives

- Clean and combine data from four publicly available climate CSV files.
- Run exploratory data analysis at the global, country, and city level.
- Engineer temporal and emissions-based features from the raw data.
- Train and compare five candidate models using TimeSeriesSplit.
- Pick the model that holds up best scientifically, not just numerically.
- Forecast global land average temperature for 2050 under a high-emissions scenario.
- Identify which countries and cities are warming fastest.

## 1.4 Scope and Limitations

This project is centred on land surface temperature at a global scale. Country- and city-level data were brought in for the exploratory work, but the final predictive model runs on 1,992 monthly observations spanning 1850 to 2015. It does not touch ocean temperatures, volcanic activity, solar

variability, or policy-driven emissions changes. Capturing any of that properly would need a much more complex multi-variable model than what we built here, and that was outside what this course project could reasonably cover.

## 2. Dataset Description

This project draws on two publicly available Kaggle datasets, five CSV files in total. Each file had a specific job in the pipeline, and not all of them ended up being used the same way.

### 2.1 Berkeley Earth Surface Temperature (BEST) Dataset

Source: [kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data](https://kaggle.com/datasets/berkeleyearth/climate-change-earth-surface-temperature-data). Compiled and maintained by Berkeley Earth. It covers land surface temperature records from 1750 to 2015. This is one of the more complete open-access historical climate records currently available, which is part of why we chose it over smaller alternatives.

| File                                  | Records  | Date Range | Purpose in Project                                   |
|---------------------------------------|----------|------------|------------------------------------------------------|
| GlobalTemperatures.csv                | 3,192    | 1750–2015  | Primary — model training and forecasting             |
| GlobalLandTemperaturesByCountry.csv   | ~577,000 | 1743–2013  | Country-level EDA and warming analysis               |
| GlobalLandTemperaturesByCity.csv      | ~8.6M    | 1743–2013  | City-level EDA, coordinate cleanup, warming analysis |
| GlobalLandTemperaturesByMajorCity.csv | —        | —          | Excluded — superseded by the city-level file         |

### 2.2 OWID CO2 and Greenhouse Gas Emissions Dataset

Source: [kaggle.com/datasets/mexwell/global-co2-and-greenhouse-gas-emissions](https://kaggle.com/datasets/mexwell/global-co2-and-greenhouse-gas-emissions). Compiled by Our World in Data. This dataset reports annual global CO2 emissions in million tonnes, broken down by country as well as world totals. We only needed the world-level column here, so that's the one we pulled out and merged with the Berkeley temperature data.

### 2.3 Final Merged Dataset

| Attribute           | Value                                                         |
|---------------------|---------------------------------------------------------------|
| Total records       | 1,992 monthly observations (expanded from 166 annual)         |
| Date range          | 1850–2015 (industrial era)                                    |
| Target variable     | LandAverageTemperature (°C, absolute)                         |
| Engineered features | co2, Month_sin, Month_cos, Temp_lag12, co2_cumulative, Decade |

| Attribute   | Value                              |
|-------------|------------------------------------|
| Merge key   | Year + Month (monthly granularity) |
| Granularity | Global, monthly average            |

One decision here mattered more than we expected going in. The first version of the merged dataset, done at yearly resolution, only had 166 records. That's simply not much to train a model on. Keeping the Berkeley data at its native monthly resolution, and expanding the OWID yearly CO2 values across all twelve months of each year, took the training set up to 1,992 records. Twelve times more data, and it showed almost immediately in how much more stable the cross-validation scores became.

### 3. Data Preprocessing and Exploratory Data Analysis

Each dataset was cleaned on its own before anything was merged. Partly this was a scale problem, and partly it was just easier to reason about one file's quirks at a time instead of trying to build one pipeline that handled everything.

#### 3.1 Approach: Clean Separately, Analyse Together

GlobalTemperatures.csv has 3,192 monthly rows. GlobalLandTemperaturesByCity.csv has more than 8.6 million. Those two numbers alone tell you why a single unified cleaning process wasn't going to work well here. Each file was handled individually, its own issues resolved first, before any aggregation or merging happened.

#### 3.2 GlobalTemperatures.csv

- Date conversion: the dt column was parsed into a Python datetime object with `pd.to_datetime()`, which let us pull Year and Month out directly.
- Missing values: filled with linear interpolation, forward- and backward-fill for the edges. Nothing was left missing afterward.
- Outlier detection: ran the IQR method, found nothing worth flagging. That's about what you'd expect from a globally averaged series — it's smooth by construction.
- Final state: 3,192 rows, 11 columns, zero missing values, zero duplicates.

#### 3.3 GlobalLandTemperaturesByCountry.csv

- Date conversion handled the same way as above.
- Missing values filled using linear interpolation within each country's own group, with forward- and backward-fill for boundary cases.

- Records were aggregated up to yearly country averages, mostly to keep the exploratory analysis manageable rather than working row by row.
- Country names were label-encoded where correlation analysis needed numeric input.

### 3.4 GlobalLandTemperaturesByCity.csv

This file gave us the most trouble, by a wide margin. Latitude and Longitude were stored as strings, not numbers. Values looked like '51.37N' and '0.73W'. You can't plot that, and you definitely can't compute a distance or a spatial trend from it directly.

- Coordinate conversion: used regular expressions to pull out the numeric part and the direction letter, then converted southward (S) and westward (W) values to negative numbers so they'd behave properly on a standard coordinate grid.
- Missing values: linear interpolation applied to the temperature columns, grouped by city.
- Aggregation: rolled up to yearly city averages, mainly to keep memory usage under control before running the warming analysis on 8.6 million rows.

### 3.5 OWID CO2 Dataset

- Filtered down to rows where country equals 'World', to isolate global totals from the country-level breakdown.
- Kept only the year and co2 columns — everything else in this dataset wasn't relevant to our merge.
- Each annual CO2 value was repeated across all twelve months of that year, which is what let us merge it with the Berkeley data at monthly resolution in the first place.
- Rows with missing CO2 values were dropped, and anything before 1850 was excluded to match the industrial-era scope of the project.

### 3.6 What the EDA Actually Showed

Global temperature trend. GlobalTemperatures.csv showed a clear upward trend across the full 1750–2015 window. Global land average temperature went from around 7.90 degrees Celsius in 1850 to about 9.83 degrees by 2015. That's a rise of roughly 1.93 degrees. What stood out more than the total change was the shape of it — the rate of increase clearly picked up after 1950, which lines up with the well-documented acceleration in industrial CO2 emissions during that same period.

Seasonal distribution. A monthly histogram of LandAverageTemperature turned out bimodal, with peaks matching Northern Hemisphere summer and winter. This was actually a fairly important finding for us, because it was strong enough to be the dominant signal in the raw data. That's the

reason we ended up encoding Month cyclically using Month\_sin and Month\_cos instead of just feeding the raw month number into a model.

CO2 and temperature relationship. Plotting CO2 against LandAverageTemperature showed a clear positive relationship. Annual global CO2 rose from around 198 million tonnes in 1850 to about 35,669 million tonnes by 2015. Temperature rose right alongside it. This is really the scientific backbone that justifies everything that came after in the modelling stage.

The city coordinate problem. The single biggest data quality issue across all four files was in the city dataset, where coordinates were stored as text instead of numbers. Without the regex fix described above, none of the spatial or geographic analysis in this report would have been possible. It's a small thing, technically, but it's the kind of thing that can quietly derail an entire analysis if it goes unnoticed.

## 4. Methodology and Feature Engineering

### 4.1 Overall Workflow

- Stage 1 — Data ingestion: all four CSV files loaded from Google Drive into Google Colab.
- Stage 2 — Individual preprocessing: each file cleaned separately, as covered in Section 3.
- Stage 3 — Monthly merging: Berkeley data kept at monthly resolution, OWID's yearly CO2 expanded to monthly, merged on Year and Month to get 1,992 monthly records.
- Stage 4 — Feature engineering: six predictive features built from the merged data.
- Stage 5 — Model training and evaluation: five candidate models trained with chronological splits and TimeSeriesSplit cross-validation.
- Stage 6 — Sequential forecasting: the 2050 prediction generated by iterating month by month from 2016 onward.

### 4.2 Feature Engineering

Our first version of the model used raw Year and Month as features. It returned an R2 of 0.9902. Encouraging at first glance, until we looked at feature importance and found that Month alone was responsible for 95.4 percent of it. That's not a climate model at that point. That's a calendar with extra steps. We scrapped that approach and built six engineered features instead.

| Feature   | Construction                          | Why It Matters                                         |
|-----------|---------------------------------------|--------------------------------------------------------|
| co2       | OWID world total (MT)                 | The physical driver of greenhouse warming              |
| Month_sin | $\sin(2\pi \times \text{Month} / 12)$ | Cyclic encoding for the Jan–Dec seasonal cycle         |
| Month_cos | $\cos(2\pi \times \text{Month} / 12)$ | Paired with Month_sin for a full cyclic representation |



| Feature        | Construction                    | Why It Matters                                              |
|----------------|---------------------------------|-------------------------------------------------------------|
| Temp_lag12     | Temperature shifted 12 steps    | Same month a year earlier — captures persistence            |
| co2_cumulative | Cumulative sum of co2           | Total CO2 emitted to date — atmospheric load                |
| Decade         | $(\text{Year} // 10) \times 10$ | A smoother decadal trend, avoids Year-CO2 multicollinearity |

### 4.3 Train–Test Split Strategy

We used a strict chronological split here, and this wasn't optional. If you split time-series data randomly, the model ends up training on records from the future while being tested on the past. That's backwards, and for a forecasting task it invalidates whatever accuracy number comes out the other end.

| Component        | Value                                                            |
|------------------|------------------------------------------------------------------|
| Training period  | 1850 to roughly 1987 (first 80% of the 1,992 monthly records)    |
| Testing period   | Roughly 1988 to 2015 (final 20% of the 1,992 monthly records)    |
| Split method     | Chronological — no data leakage                                  |
| Cross-validation | TimeSeriesSplit, 5 folds, temporal order preserved in every fold |

### 4.4 Evaluation Metrics

- $R^2$ , the coefficient of determination, tells us how much of the variance the model explains. It ranges from negative infinity up to 1.0. A negative score means the model is doing worse than just predicting the mean every time.
- MAE is the average absolute difference between predicted and actual temperature, in degrees Celsius. Easy to read, easy to explain to someone outside the project.
- RMSE works like MAE but punishes larger errors more. Useful for catching a model that's mostly fine but occasionally way off.
- CV standard deviation is the spread of scores across the TimeSeriesSplit folds. Anything under about 0.05 generally means the model is stable rather than getting lucky on one particular slice of time.

### 4.5 Technical Environment

| Component               | Detail                                                                                                                         |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Environment             | Google Colab with Google Drive integration                                                                                     |
| Language                | Python 3.x                                                                                                                     |
| Key libraries           | pandas, numpy, scikit-learn, matplotlib, seaborn, joblib                                                                       |
| Scikit-learn components | LinearRegression, Ridge, PolynomialFeatures, RandomForestRegressor, StandardScaler, TimeSeriesSplit, cross_val_score, Pipeline |

## 5. Candidate Models and Selection Rationale

We tested five models in total. Two things mattered for each one: could it explain the historical warming trend in a way that actually made scientific sense, and could it extrapolate out to a 2050 forecast at all. Some models failed on the first count, some on the second, and one failed on both.

### 5.1 Model 1 — Random Forest Regressor (Rejected)

Random Forest builds many decision trees on random subsets of the data and averages what they predict. It's usually a solid choice for non-linear relationships and tends to keep variance down. On paper this looked like a strong candidate.

| Metric           | Value        | Note                            |
|------------------|--------------|---------------------------------|
| R2 (full)        | 0.9902       | Inflated — data leakage present |
| MAE (scaled)     | 0.0654       | On scaled data, not °C          |
| Month importance | 95.37%       | Seasonal dominance only         |
| Split type       | Random 80/20 | Temporal leakage present        |

We rejected it, and there were really three separate reasons, not just one:

- Month explained 95.4% of the predictions on its own. That means the model was learning the seasonal cycle, not the warming trend underneath it.
- Tree-based models can't extrapolate past the range of their training data. A 2050 forecast simply wasn't possible with this approach, no matter how the accuracy number looked.
- The random split let future information leak into training, which inflated the R2 and made the whole result unreportable as it stood.

### 5.2 Model 2 — Linear Regression with Year and CO2 (Rejected)

This one used ordinary least squares with both Year and CO2 as predictors. The problem showed up almost immediately: Year and CO2 are correlated at  $r = 0.98$  across 1850–2015, which is about as

close to perfect multicollinearity as you can get without it actually being perfect. The model responded by assigning a negative CO<sub>2</sub> coefficient of -0.000008. That makes no physical sense whatsoever. The test-set R<sup>2</sup> came out to -1.875, which is worse than just guessing the average every time.

### 5.3 Model 3 — CO<sub>2</sub>-Only Linear Regression (Baseline)

A single-feature linear regression using just CO<sub>2</sub> as the predictor. We treated this as our baseline before moving to monthly data and the fuller feature set. It did better than we expected for something this simple, landing at R<sup>2</sup> = 0.9875 on the expanded monthly dataset.

| Metric                      | Value       | Verdict                         |
|-----------------------------|-------------|---------------------------------|
| CO <sub>2</sub> coefficient | +0.00003779 | Positive — physically correct   |
| R <sup>2</sup> (test set)   | 0.9875      | Strong baseline on monthly data |
| MAE                         | 0.3770 °C   | Slightly higher than Ridge      |
| CV mean R <sup>2</sup>      | 0.9665      | Stable (Std = 0.0488)           |

### 5.4 Model 4 — Ridge Regression (Selected)

Ridge is linear regression with an L<sub>2</sub> penalty added, which shrinks unstable coefficients instead of letting them swing wildly. That directly targets the multicollinearity problem we ran into with Model 2. On the full 1,992-record monthly dataset with all six engineered features, it beat everything else we tried, and by a fair margin on stability in particular.

| Metric                          | Value               | Verdict                              |
|---------------------------------|---------------------|--------------------------------------|
| Model                           | Ridge (alpha = 10)  | Regularised linear model             |
| R <sup>2</sup> (test set)       | 0.9927              | Highest of all models tested         |
| MAE (test set)                  | 0.2805 °C           | Low absolute error                   |
| RMSE (test set)                 | 0.3535 °C           | Within an acceptable range           |
| CV mean R <sup>2</sup> (5-fold) | 0.9919              | Consistent across every time fold    |
| CV std                          | 0.0005              | Near-zero — most stable model tested |
| 2050 forecast                   | 10.18 °C absolute   | Scientifically reasonable            |
| Warming since 1850              | +2.28 °C            | Within the IPCC range (1.5–4.5 °C)   |
| Split type                      | Chronological 80/20 | No data leakage                      |

A few things drove the decision to go with this model over the others. It had the highest accuracy and the best stability of anything we tested. Its six features cover seasonality through `Month_sin` and `Month_cos`. They cover temporal persistence through `Temp_lag12`. They cover cumulative atmospheric load through `co2_cumulative`, and the long-run decadal trend through `Decade` — each one doing a different job rather than overlapping. The L2 penalty keeps it from overfitting on a dataset of this size. It can extrapolate forward to 2050 through the sequential approach described later. And every coefficient in the model has an interpretation you can actually explain out loud, which matters more than it might sound like it does.

## 5.5 Model 5 — Polynomial Regression, Degree 2 (Rejected)

This one combined `PolynomialFeatures` at degree 2 with linear regression. The surface-level test  $R^2$  was 0.9760, which isn't bad by itself. But the `TimeSeriesSplit` cross-validation told a different story entirely. Mean  $R^2$  across folds came out to -3.2968, with a standard deviation of 7.2914. That's not a small wobble. That's the model falling apart depending on which slice of time it happened to be trained on. It was overfitting hard to local patterns in each fold and had essentially no ability to generalise across periods. We rejected it without much debate.

## 5.6 Final Model Comparison

| Model                    | Test $R^2$ | CV Mean $R^2$ | CV Std | Forecasts 2050 | Decision |
|--------------------------|------------|---------------|--------|----------------|----------|
| Random Forest            | 0.9902*    | N/A           | N/A    | No             | Rejected |
| Linear Reg. (Year + CO2) | -1.875     | N/A           | N/A    | Broken         | Rejected |
| Linear Reg. (CO2 only)   | 0.9875     | 0.9665        | 0.0488 | Yes            | Baseline |
| Ridge Regression         | 0.9927     | 0.9919        | 0.0005 | Yes            | Selected |
| Polynomial, Degree 2     | 0.9760     | -3.2968       | 7.2914 | Yes            | Rejected |

*\*Random Forest's  $R^2$  is inflated by temporal data leakage from randomly splitting time-series data.*

# 6. Results and Analysis

## 6.1 Final Model Performance

We trained Ridge on the first 80 percent of the 1,992 monthly records and tested it on the remaining 20 percent, using the strict chronological split described earlier.

| Evaluation             | R2     | MAE (°C) | RMSE (°C) | CV Mean R2 (Std) |
|------------------------|--------|----------|-----------|------------------|
| Ridge — test set       | 0.9927 | 0.2805   | 0.3535    | 0.9919 (0.0005)  |
| LR CO2-only (baseline) | 0.9875 | 0.3770   | 0.4615    | 0.9665 (0.0488)  |

Ridge beat the baseline on every single metric here. What stood out to us more than the R2 itself was the CV standard deviation of 0.0005. That's about as close to zero as you'd realistically expect from a model on real-world data, and it tells us the performance isn't tied to one lucky time window. It holds up fold after fold.

## 6.2 2050 Temperature Forecast

Because the model depends on Temp\_lag12, we couldn't just jump straight to a 2050 number in one step. Forecasting had to be sequential. We ran the model forward month by month, starting January 2016 through December 2050, feeding each month's prediction into the following year's lag value. CO2 itself was projected to grow linearly from its 2015 level up to 46,000 million tonnes by 2050.

| Reference Point             | Year | Temp (°C abs.) | Note                               |
|-----------------------------|------|----------------|------------------------------------|
| Historical baseline         | 1850 | 7.90 °C        | Start of the industrial era        |
| Last actual record          | 2015 | 9.83 °C        | End of the dataset                 |
| Ridge sequential forecast   | 2050 | 10.18 °C       | Under the 46,000 MT CO2 scenario   |
| Warming, 2015 to 2050       | —    | +0.35 °C       | Near-term additional rise          |
| Total warming, 1850 to 2050 | —    | +2.28 °C       | Within the IPCC range (1.5–4.5 °C) |

One clarification is worth making explicit here. The 10.18 degree figure is an absolute temperature reading, not an anomaly. The IPCC's commonly cited range of +1.5 to +4.5 degrees Celsius refers to change above the pre-industrial baseline, not a final reading on its own. Our model's implied warming of +2.28 degrees since 1850 falls inside that documented range. That's a reasonable sign that the forecast is at least scientifically plausible under a moderate-to-high emissions path, rather than an outlier number that would need explaining away.

## 6.3 Geographic Warming Analysis

Using the country- and city-level Berkeley data, we worked out warming trends by comparing the 1990–2000 average temperature against a 1900–1910 baseline. That's a 90-year window, and it covers most of the industrial-era warming we care about here.

| Rank | Country      | Temperature Rise |
|------|--------------|------------------|
| 1    | Mongolia     | +1.836 °C        |
| 2    | Kazakhstan   | +1.464 °C        |
| 3    | Kyrgyzstan   | +1.423 °C        |
| 4    | North Korea  | +1.337 °C        |
| 5    | Turkmenistan | +1.321 °C        |

| Rank | City             | Temperature Rise |
|------|------------------|------------------|
| 1    | Ulan Ude, Russia | +1.970 °C        |
| 2    | Chita, Russia    | +1.966 °C        |
| 3    | Irkutsk, Russia  | +1.950 °C        |
| 4    | Angarsk, Russia  | +1.950 °C        |
| 5    | Kyzyl, Russia    | +1.882 °C        |

A pattern jumps out from both tables pretty quickly. Central Asia and the Russian Far East are warming faster than almost anywhere else on record here. This isn't a coincidence, and it isn't unique to our dataset either — it lines up with the well-documented idea of continental interior amplification. Landlocked regions, without an ocean nearby to buffer temperature swings, tend to warm noticeably faster than coastal areas do. This directly answers one of the four questions we opened this report with: which regions are contributing most to the warming trend we're seeing globally.

## 6.4 Key Findings

- Global land temperatures rose by about +1.93°C between 1850 and 2015, and the rate of increase clearly sped up after 1950.
- Ridge Regression with six engineered features reached  $R^2 = 0.9927$ , with a cross-validation standard deviation of just 0.0005.
- The 2050 forecast of 10.18°C implies +2.28°C of total warming since 1850. That's consistent with what the IPCC has already projected.
- Central Asia and Russian Siberia came out as the fastest-warming regions in our analysis. Mongolia led the country-level rankings at +1.836°C.
- Polynomial Regression fell apart under TimeSeriesSplit cross-validation, with a mean  $R^2$  of -3.2968. That result alone reinforced Ridge as the right call.

- Expanding from 166 annual records to 1,992 monthly records was probably the single biggest improvement to model quality anywhere in this project.
- Coordinates stored as plain text turned out to be a fairly common data quality issue, and it needs regex-based cleaning before any spatial analysis can even start.

## 7. Conclusion and Future Work

### 7.1 Conclusion

This project set out to build a machine learning pipeline for analysing global climate change and forecasting temperature to 2050, one that could actually hold up to scrutiny rather than just post an impressive-looking accuracy number. Starting from four raw CSV files across two independent Kaggle datasets, we worked through a clean-separately-merge-together process, ran exploratory analysis at the global, country, and city level, and went through several rounds of model iteration before we landed on something we were confident in.

If we're honest, the most valuable part of this project wasn't the final R2 number. It was the methodology decisions along the way. Catching the temporal data leakage early. Rejecting a Random Forest model that looked strong on the surface but turned out to be scientifically hollow underneath. Expanding from 166 annual records to 1,992 monthly ones by preserving granularity properly instead of taking the easy route. Building six features that each had a real physical or temporal reason to exist, rather than throwing everything at the model and hoping something stuck. That combination is what let Ridge Regression reach  $R^2 = 0.9927$  with a cross-validation standard deviation of 0.0005.

The final model puts global land temperature at 10.18°C by 2050, which works out to +2.28°C of warming from the 1850 industrial baseline. The geographic analysis pointed to Central Asian countries and Russian Siberian cities as the fastest-warming places in this dataset, a finding that's consistent with established climate science and one that directly answers the question this project set out to ask in the first place.

### 7.2 Limitations

- Monthly CO2 values were approximated by splitting annual OWID totals evenly across 12 months, so real month-to-month variation in emissions wasn't captured.
- The Temp\_lag12 feature means multi-decade forecasts need sequential iteration. There's no single-step way to jump straight to a distant year.
- CO2 alone can't account for ocean heat content, solar variability, volcanic forcing, or land-use change, all of which play some role in the real climate system.

- Berkeley Earth's data stops at 2015. Bringing in more recent records would make the near-term part of the forecast considerably more useful.
- The 2050 projection assumes CO<sub>2</sub> grows linearly to 46,000 MT. Actual future emissions could easily follow a very different path from that.

### 7.3 Future Work

- Add methane, nitrous oxide, and solar irradiance as features to build a more complete multi-driver climate model.
- Bring in post-2015 NASA GISS or HadCRUT5 records to refresh and extend the training data.
- Build separate country-level regression models to look at regional vulnerability by 2050 rather than only a global figure.
- Try an LSTM neural network on the full monthly Berkeley dataset to see if it captures long-range dependencies better than Ridge does.
- Run multiple CO<sub>2</sub> emission scenarios, such as SSP1, SSP3, and SSP5, to get a range of 2050 forecasts instead of a single point estimate.

## Acknowledgements

We want to thank our professors and the senior developers who guided us through this project. Their push toward critical thinking around model selection, rather than settling for whatever number looked best first, shaped a lot of the decisions in this report.

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## Appendix: Core Code Snippets

The full notebook, including the cleaning scripts for each of the four CSV files, the feature engineering steps, the model training and evaluation code, and the sequential 2050 forecasting loop, is available in the project's GitHub repository at [github.com/binz1486/Climate-Change-And-Global-Warming-Project](https://github.com/binz1486/Climate-Change-And-Global-Warming-Project).

It wasn't included in full here to keep this report at a readable length, but every result, table, and figure in this report was generated directly from that code.

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**— *End of Report* —**